

Game Theory for Traffic Optimization

A Practical Research Topic on Multi-Agent Routing and the Price of Anarchy

Topic 2 • Game Theory & Simulation¹

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Abstract

Traffic congestion is not only a shortest-path problem; it is also a multi-agent decision-making problem in which every driver optimizes their own travel time. When too many drivers choose the same “best” route, that route becomes crowded for everyone. This document proposes a practical research project that investigates whether a game-theoretic traffic-routing algorithm can reduce overall congestion compared with selfish shortest-path routing. It outlines the background of selfish routing, Wardrop equilibrium and the Price of Anarchy, the core research question and sub-questions, a five-step research plan, the expected outcome, and source references.

1. Basic Background

Traffic congestion is not only a shortest-path problem. It is also a **multi-agent decision-making problem**. Every driver wants to choose the fastest route for themselves. However, if too many drivers choose the same “best” route, that route becomes crowded and may become slower for everyone.

This creates a classic game-theory situation called **selfish routing**. Each driver acts rationally from their own perspective, but the overall system may become inefficient. In traffic theory, this is often studied using **Wardrop equilibrium**, where no driver can improve their travel time by switching routes alone. However, this equilibrium may not be socially optimal.

The gap between selfish behavior and system-wide optimal routing is often measured by the **Price of Anarchy**. It asks: how much worse can the total traffic outcome become when every driver optimizes only for themselves?

2. Research Question

Can a game-theoretic traffic-routing algorithm reduce overall congestion compared with selfish shortest-path routing?

Possible sub-questions

1. How much worse is selfish routing compared with system-optimal routing?
2. Can incentive-based routing improve total traffic flow?
3. What happens when only some drivers follow the recommended route?
4. How does traffic demand affect congestion and equilibrium behavior?

3. Brief Research Plan

1. Build a traffic network simulation

- Model roads as a graph.
- Nodes represent intersections.

- Edges represent roads with travel-time functions.
 - Travel time increases as more cars use the same road.
2. **Implement baseline routing**
 - Selfish shortest-path routing: every driver chooses the fastest route for themselves.
 - System-optimal routing: assign drivers to minimize total travel time.
 3. **Implement game-theoretic routing**
 - Simulate drivers repeatedly choosing routes.
 - Allow route choices to converge toward equilibrium.
 - Add optional incentives, such as rewards for choosing less congested roads.
 4. **Compare routing strategies**
 - Average travel time.
 - Total system travel time.
 - Congestion on each road.
 - Fairness among drivers.
 - Stability of route choices.
 5. **Extend the study**
 - Test simple networks first, such as two parallel roads.
 - Then test a larger city-style network.
 - Study how results change as the number of drivers increases.

4. Expected Outcome

The project should demonstrate that individual shortest-path routing can lead to inefficient traffic patterns, while game-theoretic or incentive-aware routing can improve system-wide performance. This topic is strong because it combines graph algorithms, game theory, simulation, and real-world transportation problems.

References

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